



A Knowledge-Driven Framework for Conflict-Aware Evaluation of Complex Systems: An Extenics-Based Approach



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Abstract: Complex safety-critical systems are often characterised by heterogeneous knowledge representations, conflicting information sources, and inconsistent evaluation outcomes. To address these challenges, a knowledge-driven evaluation framework grounded in Extenics is developed. The framework begins with a conjugate analytical decomposition that restructures safety-related factors—personnel, equipment, environment, and management—into formally defined and comparable knowledge units, organised into a hierarchical structure comprising four primary dimensions and sixteen subordinate attributes. On this basis, matter-element models, together with classical and joint knowledge domains, are established to enable unified knowledge representation. A correlation-based reasoning mechanism is then constructed to quantify the degree of association between knowledge attributes and predefined safety states, allowing systematic treatment of contradictory and dynamically evolving information. The evaluation process is completed through weighted aggregation and the computation of extension superiority, yielding an integrated classification of system safety conditions. Application to a coal mine case using third-quarter operational data from 2024 demonstrates that the proposed framework can consistently identify overall safety levels and reveal structurally weak knowledge components. The study provides a generalisable and interpretable approach to knowledge modelling, conflict resolution, and differentiated risk governance in complex systems.

Keywords: Extenics; Knowledge representation; Knowledge conflict; Knowledge fusion; Complex systems evaluation

1 Introduction

The core of complex systems evaluation lies not only in classifying the object, but also in how to organise information from different sources, with different granularities, and even mutually conflicting, into a computable, interpretable, and inferable knowledge structure. Especially in high-risk scenarios, knowledge includes both explicit monitoring data and implicit management constraints; it reflects both safeguarding capabilities and potential vulnerabilities. Therefore, the evaluation process is essentially a process of knowledge modelling and knowledge reasoning, and its key lies in how to uniformly represent multidimensional attributes, identify conflict relationships, and describe the dynamic changes of state boundaries [1].

At present, many scholars have conducted extensive research on the evaluation and representation of complex systems. Simone et al. [2] investigated the completeness of industrial near-miss reports through a knowledge graph-based approach and proposed a structured framework for organising heterogeneous safety information. By transforming fragmented safety records into interconnected knowledge entities and relationships, the framework enabled systematic representation and analysis of complex safety information. Compared with conventional evaluation approaches relying on manually designed indicators or isolated data sources, the proposed method strengthened knowledge organisation and semantic association among system components, thereby improving information interpretability and supporting more comprehensive safety assessment processes. Zhang et al. [3] addressed the performance evaluation problem of complex systems and proposed a model based on Hierarchical Evidence Reasoning (HER) rules, which integrates multi-level uncertain information. By incorporating multi-layer indicators collected from complex systems

into a belief structure, the model can consider adaptive weights and adaptive reliability. On this basis, considering that complex systems are susceptible to internal and external disturbances leading to performance fluctuations, the study introduced a disturbance analysis method based on HER rules to quantify the impact of disturbances on system performance. Finally, a load-bearing system was used as a case to verify the rationality of the HER rules and evaluate the influence of disturbances on overall system performance. Mulumba et al. [4] established an underground coal mine safety risk evaluation model based on an optimised Particle Swarm Optimisation-Back Propagation (PSO-BP) neural network. Zhang et al. [5] proposed a new complex system health condition evaluation method—interpretable reference value optimisation Belief Rule Base (I-BRB). To address inaccurate reference values, the study designed a reference value optimisation algorithm with interpretability constraints, which optimises reference values without compromising expert knowledge. Subsequently, a Projection Covariance Matrix Adaptation Evolution Strategy (P-CMA-ES) combined with interpretability constraints was employed to optimise the remaining parameters and improve model accuracy. Finally, a case study evaluating bearing components in a flywheel system verified the effectiveness of the proposed method. Experimental results showed that I-BRB achieved higher accuracy in health condition evaluation compared with other methods. Cai et al. [6] proposed a knowledge graph-driven fault diagnosis method for intelligent manufacturing systems. The proposed framework constructs a multi-level knowledge graph by integrating heterogeneous information from equipment states, fault characteristics, and operational data, enabling structured representation of fault knowledge. In addition, knowledge reasoning mechanisms were introduced to identify relationships among fault events and system components, thereby improving the interpretability of the diagnosis process. A case study involving rotating machinery fault diagnosis was conducted to verify the effectiveness of the proposed approach. The results showed that the model improves fault identification performance and captures complex interactions within industrial systems more effectively than conventional methods. Eshwar et al. [7] proposed an Information Entropy-based Risk (IER) index for assessing safety risks in underground mines. To address the limitations of traditional MSHA tools such as the Pattern of Violation (POV) and Significant & Substantial (S&S) calculator, the study integrated multiple risk-related indicators from MSHA databases into a unified evaluation framework. Seven risk indicators were considered, including citations, orders, significant and substantial citations, penalties, incidents with no lost time, lost-time injuries, and proposed penalties for violations. An information entropy method was employed to optimise the weights of these potentially conflicting indicators and generate a comprehensive risk index. Subsequently, the BIRCH clustering algorithm, combined with statistical analyses such as MANOVA, post hoc tests, box plots, and ANOVA, was used to validate the effectiveness of the proposed index. The results showed that the IER index could effectively distinguish different mine safety risk levels, with statistically significant differences observed among clusters. Finally, the method was applied to an underground coal mine case, demonstrating its practical value in supporting safety performance assessment and risk management in mining operations. Tripathy et al. [8] classified hazards occurring in different parts of underground mining and predicted related risks using different machine learning modelling techniques such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), logistic regression, and decision trees. The method followed the regulatory guidelines of Directorate General of Mines Safety (DGMS) and used them as the basic building modules for constructing machine learning classification models. Although current complex systems evaluation methods have made certain progress, several obvious limitations remain. First, most existing studies focus on the integration of a single type of information or knowledge source, lacking effective multi-source heterogeneous knowledge fusion mechanisms, which leads to limitations in coping with complex dynamic changes of systems. Second, many methods lack effective mechanisms for handling conflicts between implicit and explicit knowledge, making it difficult for the evaluation process to comprehensively and accurately characterise system states. Particularly in high-risk scenarios, nonlinear and disturbance effects of the system are not sufficiently considered. In addition, current evaluation models mostly focus on static characteristics and ignore uncertainties and external disturbances during dynamic system evolution, making it difficult to reflect system changes in real time and resulting in poor model adaptability. Although some optimisation algorithms improve accuracy, they often sacrifice interpretability. However, in complex systems evaluation, decision-makers require not only accurate results but also an understanding of the reasoning process behind the conclusions. Therefore, how to improve model accuracy while maintaining interpretability and effectively coping with dynamic changes remains a major challenge in current research.

To address the above issues, this paper proposes a knowledge-driven evaluation framework based on Extenics theory. The framework effectively integrates multidimensional safety attributes from different sources and combines a reasoning mechanism under dynamic environments to identify and quantify potential risks in the system [9]. Specifically, the contributions of this study are as follows. First, a knowledge representation framework based on Extenics theory is proposed to systematically characterise multidimensional safety attributes in complex systems. By constructing an adaptive knowledge structure, different types of safety elements can be effectively represented within a unified framework. Second, a knowledge decomposition mechanism based on conjugate analysis is constructed to achieve unified modelling of implicit and explicit knowledge, successfully resolving conflicts arising during multi-source knowledge representation and reasoning. Finally, a knowledge reasoning method based on

correlation functions is proposed. By quantifying the relationships among knowledge attributes, contradictions and inconsistencies among indicators are effectively handled, thereby providing an interpretable integrated classification result for system safety states [10]. Through a case study in a coal mine safety scenario, this paper not only demonstrates the practical application potential of the framework but also highlights its scalability as a general method. The framework provides new theoretical support for risk cognition, knowledge modelling, and safety governance in complex systems, and has important academic value and application prospects.

2 Knowledge Structure Construction of Complex Safety Systems Based on Conjugate Analysis

The coal mine safety system can be regarded as a typical complex knowledge system, whose state is not determined by a single indicator, but is jointly formed by multiple types of knowledge such as personnel, equipment, environment, and management [11]. Based on the conjugate analysis theory of Extenics, this paper transforms the traditional “evaluation indicator selection” process into a “knowledge unit identification and knowledge structure construction” process: by identifying the positive–negative parts, latent–manifest parts, soft–hard parts, and virtual–real parts in the system, key knowledge attributes that can characterise safety capability, risk sources, potential vulnerabilities, and governance constraints are extracted, and organised into a hierarchical knowledge structure [12].

The positive–negative conjugate dimension focuses on the unity of opposites between safeguarding factors and risk factors in safety elements, and selects indicators that are quantifiable, highly dynamic, and capable of clearly defining critical points. In terms of personnel safety, the certified operation rate (safeguarding factor) and violation operation frequency (risk source) are selected, which can effectively and dynamically characterise the balancing process between safety capability support and violation impact. For equipment safety, equipment integrity rate (safeguard) and main ventilation fan failure rate (risk) are adopted. For environmental safety, gas concentration mean (risk) and dust concentration compliance rate (safeguard) are taken as the core indicators. For management safety, the safety management system implementation rate and safety hazard rectification rate are selected as two positive safeguarding indicators, reflecting the ability of the management system to suppress risks.

The latent–manifest conjugate dimension focuses on the transformation relationship between implicit causes and explicit events, and gives priority to latent indicators with predictive capability. In terms of personnel safety, employee safety assessment pass rate (latent quality) is retained, while accident frequency is excluded due to strong lagging characteristics. For equipment safety, monitoring system effectiveness rate (latent barrier) is selected. For environmental safety, roof support quality compliance rate (latent control) is taken as a key indicator. For management safety, safety management system implementation rate (latent prevention) is retained, while explicit risks are reflected by indicators in other dimensions.

The soft–hard conjugate dimension coordinates the interaction efficiency between soft constraints such as management culture and physical entity states. In terms of personnel safety, cultural identity with strong subjectivity is excluded, and quantified training indicators are adopted. For equipment safety, equipment integrity rate and similar indicators reflect physical states, while soft management effects are incorporated into the system implementation rate. For environmental safety, hard indicators such as gas concentration mean are mainly used, and the effects of soft measures are reflected through management indicators. For management safety, emergency drill frequency and safety investment proportion are taken as soft indicators, while static geological attribute indicators are excluded due to lack of dynamic influence.

The virtual–real conjugate dimension emphasises that abstract safety capability is mapped to measurable data entities. In terms of personnel safety, personnel training rate is used as a real-part indicator. For equipment safety, real-part data such as equipment integrity rate directly quantify virtual-part capabilities such as reliability. For environmental safety, real-part indicators such as drainage system reliability map disaster resistance capability. For management safety, safety investment proportion (virtual part) and safety hazard rectification rate (real part) jointly reflect management effectiveness, while decision efficiency is excluded due to difficulty in quantification.

Based on the above analysis, this paper constructs a knowledge attribute screening basis table for complex safety systems, as shown in Table 1.

Table 1. Knowledge attribute screening basis table for complex safety systems

	Positive–Negative	Latent–Manifest	Soft–Hard	Virtual–Real	Final Selected Indicators
Personnel safety	Employee satisfaction Psychological fatigue index Certified operation rate Violation operation frequency	Employee safety assessment pass rate	Enterprise culture identity	Personnel training rate	Personnel training rate Certified operation rate Employee safety assessment pass rate Violation operation frequency

Equipment safety	Equipment integrity rate Main ventilation fan failure rate	Monitoring system effectiveness rate Equipment design defect rate	Equipment integrity rate Main ventilation fan failure rate Maintenance procedure implementation rate	Equipment integrity rate Main ventilation fan failure rate Monitoring system effectiveness rate Spare parts inventory turnover rate	Equipment integrity rate Main ventilation fan failure rate Safety protection device configuration rate Monitoring system effectiveness rate
Operation safety	Gas concentration mean Dust concentration compliance rate	Roof support quality compliance rate Real-time roof displacement	Gas concentration mean Dust concentration compliance rate Drainage system reliability Dust removal system	Drainage system reliability Gas concentration mean Dust concentration compliance rate	Gas concentration mean Dust concentration compliance rate Roof support quality compliance rate Drainage system reliability
Management safety	Safety management system implementation rate Safety hazard rectification rate	Safety management system implementation rate	Emergency drill frequency Safety investment proportion Roadway geological structure complexity	Safety investment proportion Safety hazard rectification rate Management decision efficiency	Safety management system implementation rate Emergency drill frequency Safety hazard rectification rate Safety investment proportion

Based on Table 1, this paper performs merging, redundancy removal, and consistency processing on candidate knowledge attributes, and finally forms a knowledge structure composed of four primary knowledge dimensions and sixteen secondary knowledge attributes, as shown in Figure 1.

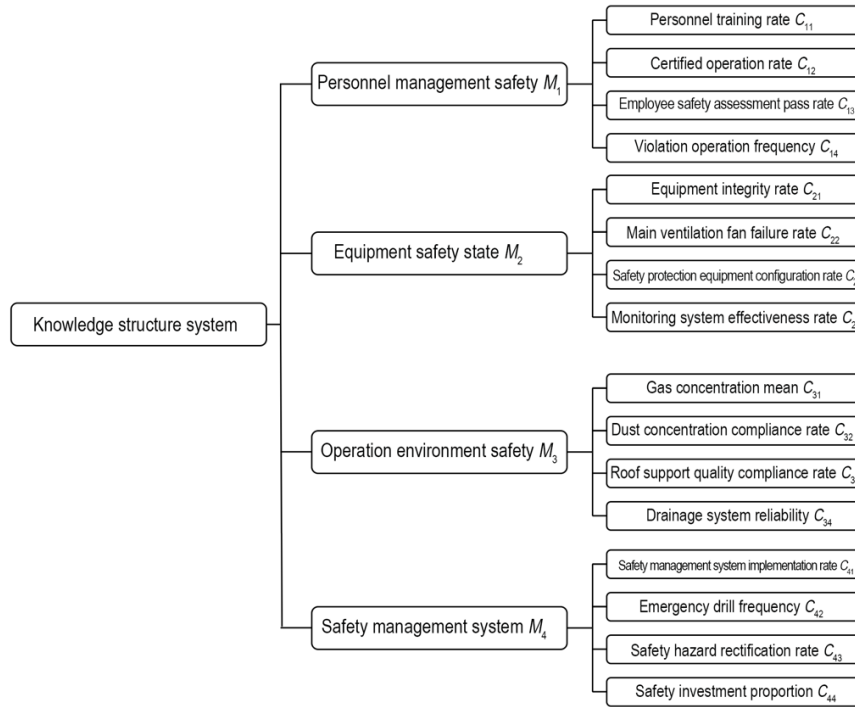


Figure 1. Knowledge structure system

3 Knowledge-Driven Extenics Reasoning Model

3.1 Knowledge Attribute Category Analysis

Correlation analysis of the indicators in the above-established knowledge structure system shows that some evaluation indicators, under the influence of affecting factors, usually exhibit steady-state or slowly varying characteristics, and can be effectively represented by a single deterministic statistical value. However, some evaluation indicators, under the influence of affecting factors, show non-stationary fluctuations within the monitoring period, and are difficult to be accurately represented by a single static value, thus possessing fuzziness [13]. Therefore, the evaluation indicators in the coal mine evaluation indicator system are divided into two categories: static indicators and dynamic indicators.

Static indicators generally show high correlation stability and can be measured by specific numerical values. Specifically, the boundary between the positive-part safeguarding factors and negative-part risk factors represented by such indicators is relatively clear, and fundamental reversal is unlikely to occur under routine system operation fluctuations. The transformation process from latent-part implicit causes to manifest-part events usually presents gradual characteristics, or mainly depends on the continuous action of institutionalised and routine management behaviours. Meanwhile, the interaction efficiency between soft-part constraints such as management culture and hard-part physical entity states, as well as the mapping relationship from virtual-part abstract safety capability to real-part measurable data, both exhibit relatively stable and predictable characteristics. In the personnel safety management evaluation module, personnel training rate, certified operation rate, employee safety assessment pass rate, and violation operation frequency are static indicators. In the equipment safety state evaluation module, equipment integrity rate, main ventilation fan failure rate, safety protection equipment configuration rate, and monitoring system effectiveness rate are static indicators. In the operation environment safety evaluation module, roof support quality compliance rate is a static indicator. In the safety management system evaluation module, safety management system implementation rate, emergency drill frequency, safety hazard rectification rate, and safety investment proportion are static indicators.

The core characteristics of dynamic indicators lie in their significant correlation variability and high sensitivity to dynamic factors of the production environment, and they can be measured using dynamic fuzzy values. In the positive–negative dimension, the safety states represented by such indicators are often located in critical fuzzy regions, where slight environmental disturbances may induce qualitative change toward dangerous states. In the latent–manifest dimension, the associated deep implicit risks (such as concealed geological anomalies or potential abnormal gas accumulation) tend to rapidly transform into explicit disaster events. In the soft–hard dimension, their safety effectiveness highly depends on real-time monitoring and feedback of physical entity states. In the virtual–real dimension, although the theoretical mapping relationship between abstract risk concepts and specific monitoring data is clear, the monitoring data itself, as the real-part representation, exhibits strong spatiotemporal non-uniform fluctuation characteristics. For example, gas concentration mean, as a typical instance of special indicators, fluctuates in real time with changes in underground mining activities and ventilation efficiency [14]. The dust concentration compliance rate is affected by spatiotemporal factors such as spatial distribution of working faces and equipment utilisation rate, and also exhibits dynamic variation characteristics. The drainage system reliability changes dynamically with seasonal variations in hydrological conditions, and its bearing capacity presents dynamic variation [15]. When evaluating such indicators, the data usually need to be presented in the form of specific fluctuation intervals. In the operation environment safety evaluation module, gas concentration mean, dust concentration compliance rate, and drainage system reliability are dynamic indicators. In recent years, data-driven coal mine environment safety risk evaluation studies have further shown that dynamic monitoring data can be used to identify environmental risks in real time and support key risk control [16].

3.2 Matter-Element Models of Evaluation Indicator Joint Domain and Classical Domain

For the various knowledge attributes in the above knowledge structure, the matter-element representation model of the evaluation object is constructed based on Extenics theory. The matter-element model of the personnel management safety evaluation module is expressed as:

$$R_1 = \begin{bmatrix} M_1 & C_{11} & v_{11} \\ & C_{12} & v_{12} \\ & C_{13} & v_{13} \\ & C_{14} & v_{14} \end{bmatrix} \quad (1)$$

where, M_1 represents the personnel management safety evaluation module; $C_{11}, C_{12}, C_{13}, C_{14}$ represent the indicators personnel training rate, certified operation rate, employee safety assessment pass rate, and violation operation frequency, respectively; $v_{11}, v_{12}, v_{13}, v_{14}$ represent the values of evaluation indicators $C_{11}, C_{12}, C_{13}, C_{14}$, respectively.

The matter-element model of the equipment safety state evaluation module is expressed as:

$$R_2 = \begin{bmatrix} M_2 & C_{21} & v_{21} \\ & C_{22} & v_{22} \\ & C_{23} & v_{23} \\ & C_{24} & v_{24} \end{bmatrix} \quad (2)$$

where, M_2 represents the equipment safety state evaluation module; $C_{21}, C_{22}, C_{23}, C_{24}$ represent the indicators equipment integrity rate, main ventilation fan failure rate, safety protection equipment configuration rate, and monitoring system effectiveness rate, respectively; $v_{21}, v_{22}, v_{23}, v_{24}$ represent the values of evaluation indicators $C_{21}, C_{22}, C_{23}, C_{24}$, respectively.

The matter-element model of the operation environment safety evaluation module is expressed as:

$$R_3 = \begin{bmatrix} M_3 & C_{31} & (a_{31}, b_{31}) \\ & C_{32} & (a_{32}, b_{32}) \\ & C_{33} & v_{33} \\ & C_{34} & (a_{34}, b_{34}) \end{bmatrix} \quad (3)$$

where, M_3 represents the operation environment safety evaluation module; C_{31} , C_{32} , C_{33} , C_{34} represent the indicators gas concentration mean, dust concentration compliance rate, roof support quality compliance rate, and drainage system reliability, respectively; v_{33} represents the value of evaluation indicator C_{33} ; and (a_{31}, b_{31}) , (a_{32}, b_{32}) , (a_{34}, b_{34}) represent the dynamic variation intervals of evaluation indicators C_{31} , C_{32} , C_{34} , respectively.

The matter-element model of the safety management system evaluation module is expressed as:

$$R_4 = \begin{bmatrix} M_4 & C_{41} & v_{41} \\ & C_{42} & v_{42} \\ & C_{43} & v_{43} \\ & C_{44} & v_{44} \end{bmatrix} \quad (4)$$

where, M_4 represents the safety management system evaluation module; C_{41} , C_{42} , C_{43} , C_{44} represent safety management system implementation rate, emergency drill frequency, safety hazard rectification rate, and safety investment proportion, respectively; v_{41} , v_{42} , v_{43} , v_{44} represent the values of evaluation indicators C_{41} , C_{42} , C_{43} , C_{44} , respectively.

The matter-element model of the evaluation system is expressed as:

$$R = \begin{bmatrix} S & C_1 & v_1 \\ & C_2 & v_2 \\ & C_3 & v_3 \\ & C_4 & v_4 \end{bmatrix} \quad (5)$$

where, S represents the coal mine safety grade; C_1 , C_2 , C_3 , C_4 represent the primary indicators personnel management safety, equipment safety state, operation environment safety, and safety management system, respectively; v_1 , v_2 , v_3 , v_4 represent the values of evaluation indicators C_1 , C_2 , C_3 , C_4 , respectively.

Correspondingly, the classical domain matter-element model of evaluation indicators is:

$$R_j = (Z_j, C, V_j) = \begin{bmatrix} Z_j & C_{11} & V_{11} \\ & C_{12} & V_{12} \\ & \vdots & \vdots \\ & C_{44} & V_{44} \end{bmatrix} = \begin{bmatrix} Z_j & C_{11} & (a_{11}, b_{11}) \\ & C_{12} & (a_{12}, b_{12}) \\ & \vdots & \vdots \\ & C_{44} & (a_{44}, b_{44}) \end{bmatrix} \quad (6)$$

where, Z_j represents the j -th risk grade, and $V_{11}, V_{12}, \dots, V_{44}$ represent the value ranges specified by Z_j for the evaluation indicators $C_{11}, C_{12}, \dots, C_{44}$, namely the classical domains, with $V_{ij} = (a_{ij}, b_{ij})$.

The joint domain matter-element model of evaluation indicators is:

$$R_p = (P, C, V_p) = \begin{bmatrix} P & C_{11} & V_{11p} \\ & C_{12} & V_{12p} \\ & \vdots & \vdots \\ & C_{44} & V_{44p} \end{bmatrix} = \begin{bmatrix} P & C_{11} & (a_{11p}, b_{11p}) \\ & C_{12} & (a_{12p}, b_{12p}) \\ & \vdots & \vdots \\ & C_{44} & (a_{44p}, b_{44p}) \end{bmatrix} \quad (7)$$

where, P represents the whole set of coal mine safety, and $V_{11p}, V_{12p}, \dots, V_{44p}$ represent the value ranges taken by P for the evaluation indicators $C_{11}, C_{12}, \dots, C_{44}$.

3.3 Matter-Element Models of Evaluation Indicator Joint Domain and Classical Domain

Different knowledge attributes do not have the same importance to the integrated classification results; therefore, knowledge importance weights need to be assigned. The specific processing method is as follows.

For indispensable indicators, an index Λ is used to represent them; for other evaluation indicators, values within $[0, 1]$ are assigned according to their importance degree. The weight is denoted as:

$$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$$

where, if $\alpha_{i_0} = \Lambda$, then $\sum_{\substack{k=1 \\ k \neq i_0}}^n \alpha_k = 1$.

To reduce subjective arbitrariness, this paper adopts the Analytic Hierarchy Process (AHP) to determine the relative importance of knowledge attributes. A judgement matrix is constructed through pairwise comparison, and the weight results are obtained after the consistency test ($CI < 0.1$ and $CR < 0.1$) is satisfied [17].

3.4 Establishment of Knowledge Reasoning Mechanism

To quantify the matching degree between each knowledge attribute and different knowledge classifications, a knowledge reasoning mechanism based on extension sets is established. By calculating the distance between the object knowledge attributes and the classical domain and joint domain, it is determined which safety state it is closer to and the degree of deviation. That is:

$$K_j(v_i) = \begin{cases} -\frac{\rho(v_i, V_{ij})}{|V_{ij}|}, & v_i \in V_{ij} \\ \frac{\rho(v_i, V_{ij})}{\rho(v_i, V_{iv}) - \rho(v_i, V_{ii})}, & v_i \notin V_{ij} \end{cases} \quad (8)$$

where,

$$\rho(v_i, V_{ij}) = \left| v_i - \frac{a_{ij} + b_{ij}}{2} \right| - (b_{ij} - a_{ij})/2, \quad |V_{ij}| = |a_{ij} - b_{ij}|,$$

$$\rho(v_i, V_{ip}) = \left| v_i - \frac{a_{ip} + b_{ip}}{2} \right| - (b_{ip} - a_{ip})/2. \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, m).$$

Since the joint domain range of some knowledge attributes is a one-direction infinite interval $(\alpha, +\infty)$, in order to adapt to this situation and ensure the rationality of extension distance calculation, the extension distance is redefined as:

$$\rho(v_i, V_{ip}) = \alpha - v_i \cdot \rho(v_i, V_{ij}) = \left| v_i - \frac{c_j + d_j}{2} \right| - \frac{d_j - c_j}{2}.$$

In the formula: $K_j(v_i)$ represents the knowledge reasoning mechanism, indicating the degree to which v_i belongs to V_{ij} ; $\rho(v_i, V_{ij})$ represents the extension distance from v_i to V_{ip} ; $|V_{ij}|$ represents the absolute value of classical domain V_{ij} .

In the coal mine case, knowledge attributes such as gas concentration mean, dust concentration compliance rate, and drainage system reliability show obvious fluctuation. To ensure the scientific nature of evaluation results, the critical values of indicators are used for reasoning when calculating the correlation degree, so as to better reflect the knowledge state and risk boundary under the most unfavourable working conditions.

3.5 Normalised Knowledge Correlation Degree Calculation

The correlation degree obtained from the knowledge reasoning mechanism can be understood as the matching strength between knowledge attributes and target knowledge classifications. The larger the correlation degree value, the closer the corresponding knowledge attribute is to the standard corresponding to the classification; the smaller the correlation degree, the farther it deviates from the target classification.

To obtain comparable integrated results, it is necessary to further normalise the correlation degrees of each knowledge attribute with respect to different knowledge classifications, and obtain the normalised knowledge correlation degree k_{ij} . The calculation formula is as follows:

$$k_{ij} = \frac{K_j(v_i)}{\max |K_j(v)|} \quad (9)$$

where, $K_j(v_i)$ represents the correlation degree value in the i -th row and j -th column; $\max |K_j(v)|$ represents the maximum correlation degree value in the j -th column.

3.6 Integrated Knowledge Classification Superiority Calculation

According to the correlation degree values of each knowledge attribute with respect to different knowledge classifications and their knowledge importance weights, the integrated knowledge classification superiority can be further calculated. According to the formula:

$$c(Z_j) = \sum_{i=1}^n a_i k_{ij} \quad (10)$$

The superiority is calculated, and the superiority of Z_j is compared: if $C(Z_0) = \max_{j \in \{1, 2, \dots, m\}} \{C(Z_j)\}$, then object Z_0 is better.

4 Case Study: Framework Validation in A Coal Mine Scenario

To verify the applicability of the proposed knowledge-driven framework in complex safety systems, this paper conducts an empirical analysis using a coal mine in North China as a case study. The case retains coal mine domain characteristics, but its role is framework validation rather than the only limitation of method applicability. By mapping mine safety data into knowledge attributes and introducing classical domains and joint domains for reasoning, the effectiveness of the framework in multi-source knowledge fusion, conflict handling, and integrated classification can be examined.

4.1 Basic Data Collection and Preprocessing

Safety production data of the coal mine in the third quarter of 2024 are selected as samples. The initial data of 16 secondary indicators are obtained through on-site monitoring, ledger review, and employee interviews. Considering the timeliness and dynamics of the data, quarterly averages are adopted for continuous monitoring indicators (such as gas concentration and dust concentration), and quarterly cumulative values are taken for statistical indicators (such as training rate and failure rate). The measured values of each indicator are shown in Table 2.

Table 2. Measured values of each indicator

Primary Indicator	Secondary Indicator	Unit	Measured Value
Personnel safety management (M_1)	Personnel training rate (C_{11})	%	97.5
	Certified operation rate (C_{12})	%	100
	Employee safety assessment pass rate (C_{13})	%	94.2
	Violation operation frequency (C_{14})	times/month	2
Equipment safety state (M_2)	Equipment integrity rate (C_{21})	%	95.8
	Main ventilation fan failure rate (C_{22})	%	0.4
	Safety protection equipment configuration rate (C_{23})	%	99.2
	Monitoring system effectiveness rate (C_{24})	%	98.7
Operation environment safety (M_3)	Gas concentration mean (C_{31})	volume percentage %	0.2~0.5
	Dust concentration compliance rate (C_{32})	%	88.4~92.8
	Roof support quality compliance rate (C_{33})	%	96.7
	Drainage system reliability (C_{34})	%	87.6~92.3
Safety management system (M_4)	Safety management system implementation rate (C_{41})	%	94.6
	Safety hazard rectification rate (C_{42})	%	100
	Emergency drill frequency (C_{43})	times/quarter	2
	Safety investment proportion (C_{44})	%	5.2

4.2 Establishment of Classical Domain, Joint Domain, and Matter-Element to Be Evaluated

According to the Coal Mine Safety Regulations (2022 edition), GB 50215-2015 Code for Design of Coal Mine Engineering, and other standards, the coal mine safety state is divided into five grades (Grade I—safe, Grade II—relatively safe, Grade III—general, Grade IV—relatively dangerous, Grade V—dangerous). The classical domain matter-element matrix of the 16 secondary indicators under each grade is constructed as follows:

$$R_j = \begin{bmatrix} & Z_1 & Z_2 & Z_3 & Z_4 & Z_5 \\ C_{11} & (95, 100] & (90, 95] & (85, 90] & (80, 85] & (0, 80] \\ C_{12} & (98, 100] & (95, 98] & (90, 95] & (85, 90] & (0, 85] \\ C_{13} & (95, 100] & (90, 95] & (85, 90] & (80, 85] & (0, 80] \\ C_{14} & (0, 1] & (1, 3] & (3, 5] & (5, 8] & (8, +\infty) \\ C_{21} & (95, 100] & (90, 95] & (85, 90] & (80, 85] & (0, 80] \\ C_{22} & (0, 0.5] & (0.5, 1] & (1, 2] & (2, 3] & (3, +\infty) \\ C_{23} & (95, 100] & (90, 95] & (85, 90] & (80, 85] & (0, 80] \\ C_{24} & (98, 100] & (95, 98] & (90, 95] & (85, 90] & (0, 85] \\ C_{31} & (0, 0.3] & (0.3, 0.5] & (0.5, 0.65] & (0.65, 0.75] & (0.75, +\infty) \\ C_{32} & (95, 100] & (90, 95] & (85, 90] & (80, 85] & (0, 80] \\ C_{33} & (95, 100] & (90, 95] & (85, 90] & (80, 85] & (0, 80] \\ C_{34} & (90, 100] & (80, 90] & (70, 80] & (60, 70] & (0, 60] \\ C_{41} & (95, 100] & (90, 95] & (85, 90] & (80, 85] & (0, 80] \\ C_{42} & (98, 100] & (95, 98] & (90, 95] & (85, 90] & (0, 85] \\ C_{43} & (4, +\infty) & (3, 4] & (2, 3] & (1, 2] & (0, 1] \\ C_{44} & (5, +\infty) & (4, 5] & (3, 4] & (2, 3] & (0, 2] \end{bmatrix}$$

According to the classical domain, the joint domain is expressed as:

$$R_P = \begin{bmatrix} P & C_{11} & [0, 100] \\ & C_{12} & [0, 100] \\ & C_{13} & [0, 100] \\ & C_{14} & (0, +\infty) \\ & C_{21} & [0, 100] \\ & C_{22} & [0, 100] \\ & C_{23} & [0, 100] \\ & C_{24} & [0, 100] \\ & C_{31} & [0, 100] \\ & C_{32} & [0, 100] \\ & C_{33} & [0, 100] \\ & C_{34} & [0, 100] \\ & C_{41} & [0, 100] \\ & C_{42} & [0, 100] \\ & C_{43} & (0, +\infty) \\ & C_{44} & [0, 100] \end{bmatrix}$$

The matter-element model is:

$$R = \begin{bmatrix} S & C_{11} & 97.5 \\ & C_{12} & 100 \\ & C_{13} & 94.2 \\ & C_{14} & 2 \\ & C_{21} & 95.8 \\ & C_{22} & 0.4 \\ & C_{23} & 99.2 \\ & C_{24} & 98.7 \\ & C_{31} & 0.2 \sim 0.5 \\ & C_{32} & 88.4 \sim 92.8 \\ & C_{33} & 96.7 \\ & C_{34} & 87.6 \sim 92.3 \\ & C_{41} & 94.6 \\ & C_{42} & 100 \\ & C_{43} & 2 \\ & C_{44} & 5.2 \end{bmatrix}$$

4.3 Calculation of Knowledge Importance Weights

The AHP is adopted to determine the weights of knowledge attributes, and the relative importance of each knowledge attribute is calculated by constructing a judgement matrix. Experts in the coal mine safety field are invited to conduct pairwise comparison of the importance of four primary indicators (personnel safety management, equipment safety state, operation environment safety, and safety management system) and 16 secondary indicators, construct the judgement matrix, and perform consistency testing. Taking the primary indicators as an example, the judgement matrix is constructed as follows:

$$\begin{bmatrix} 1 & 2 & 1/2 & 3 \\ 1/2 & 1 & 1/3 & 2 \\ 2 & 3 & 1 & 4 \\ 1/3 & 1/2 & 1/4 & 1 \end{bmatrix}$$

The weights are calculated by the eigenvalue method. After the consistency test ($CI = 0.0234 < 0.1$, $CR = 0.0263 < 0.1$), the weights of the primary indicators are determined as: $\alpha_{C_1} = 0.223$, $\alpha_{C_2} = 0.132$, $\alpha_{C_3} = 0.432$, $\alpha_{C_4} = 0.213$. Similarly, the weights of the secondary indicators can be obtained. The final results are shown in Table 3.

Table 3. Weights of evaluation indicators

Primary Indicator	Weight	Secondary Indicator	Weight
M_1	0.223	C_{11}	0.292
		C_{12}	0.268
		C_{13}	0.255
		C_{14}	0.185
M_2	0.132	C_{21}	0.347
		C_{22}	0.303
		C_{23}	0.198
		C_{24}	0.152

M_3	0.432	C_{31}	0.308
		C_{32}	0.273
		C_{33}	0.247
		C_{34}	0.172
M_4	0.213	C_{41}	0.303
		C_{42}	0.288
		C_{43}	0.247
		C_{44}	0.162

4.4 Evaluation Results and Analysis

Substituting the actual values of knowledge attributes to be evaluated into Eq. (1) and Eq. (2), the correlation degrees and normalised correlation degrees of each knowledge attribute in personnel safety management, equipment safety state, operation environment safety, and safety management system are obtained. Finally, the integrated superiority is calculated according to Eq. (10). The reasoning results of the personnel safety management dimension are shown in Table 4.

Table 4. Personnel safety management superiority reasoning table

Indicator	Weight	Correlation Degree					Normalised Correlation Degree				
		Z_1	Z_2	Z_3	Z_4	Z_5	Z_1	Z_2	Z_3	Z_4	Z_5
C_{11}	0.292	0.5	-0.5	-0.75	-0.833	-0.875	1	-0.5	-0.75	-0.833	-0.875
C_{12}	0.268	0	-1	-1	-1	-1	0	-1	-1	-1	-1
C_{13}	0.255	-0.121	0.16	-0.42	-0.613	-0.71	-0.242	0.16	-0.42	-0.613	-0.71
C_{14}	0.185	-0.333	0.5	-0.333	-0.6	-0.75	-0.666	0.5	-0.333	-0.6	-0.75
Extension superiority							0.107	-0.281	-0.656	-0.779	-0.843

The reasoning results of the equipment safety state dimension are shown in Table 5.

Table 5. Equipment safety state superiority reasoning table

Indicator	Weight	Correlation Degree					Normalised Correlation Degree				
		Z_1	Z_2	Z_3	Z_4	Z_5	Z_1	Z_2	Z_3	Z_4	Z_5
C_{21}	0.347	0.16	-0.16	-0.58	-0.72	-0.79	0.8	-0.191	-0.63	-0.76	-0.823
C_{22}	0.303	0.2	-0.2	-0.6	-0.8	-0.867	1	-0.238	-0.652	-0.845	-0.903
C_{23}	0.198	0.16	-0.84	-0.92	-0.947	-0.96	0.8	-1	-1	-1	-1
C_{24}	0.152	0.15	-0.35	-0.74	-0.87	-0.913	0.75	-0.417	-0.804	-0.919	-0.951
Extension superiority							0.853	-0.4	-0.737	-0.857	-0.902

The reasoning results of the operation environment safety dimension are shown in Table 6.

Table 6. Operation environment safety superiority evaluation table

Indicator	Weight	Correlation Degree					Normalised Correlation Degree				
		Z_1	Z_2	Z_3	Z_4	Z_5	Z_1	Z_2	Z_3	Z_4	Z_5
C_{31}	0.308	-0.286	0	0	-0.231	-0.333	-0.788	0	0	-0.296	-0.399
C_{32}	0.273	-0.363	-0.121	0.32	-0.227	-0.42	-1	-0.357	0.478	-0.291	-0.503
C_{33}	0.247	0.36	-0.34	-0.67	-0.78	-0.835	0.993	-1	-1	-1	-1
C_{34}	0.172	-0.162	0.24	-0.38	-0.587	-0.69	-0.447	0.706	-0.567	-0.752	-0.826
Extension superiority							-0.387	-0.223	-0.214	-0.547	-0.649

The reasoning results of the safety management system dimension are shown in Table 7.

Table 7. Safety management system superiority evaluation table

Indicator	Weight	Correlation Degree					Normalised Correlation Degree				
		Z_1	Z_2	Z_3	Z_4	Z_5	Z_1	Z_2	Z_3	Z_4	Z_5
C_{41}	0.303	-0.069	0.08	-0.46	-0.64	-0.73	-1	0.08	-0.46	-0.64	-0.73
C_{42}	0.288	0	-1	-1	-1	-1	0	-1	-1	-1	-1
C_{43}	0.247	0	-0.333	0	0	-0.5	0	-0.333	0	0	-0.5
C_{44}	0.162	0.002	-0.004	-0.023	-0.042	-0.061	0.029	-0.004	-0.023	-0.042	-0.061

Extension superiority	-0.298	-0.347	-0.431	-0.489	-0.643
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Based on the above data, the superiority of the coal mine safety evaluation results can be calculated, as shown in Table 8.

Table 8. Coal mine safety grade superiority evaluation table

Indicator	Weight	Correlation Degree					Normalised Correlation Degree				
		Z ₁	Z ₂	Z ₃	Z ₄	Z ₅	Z ₁	Z ₂	Z ₃	Z ₄	Z ₅
M ₁	0.223	0.054	-0.281	-0.656	-0.779	-0.843	0.107	-0.281	-0.656	-0.779	-0.843
M ₂	0.132	0.171	-0.336	-0.678	-0.812	-0.866	0.853	-0.4	-0.737	-0.858	-0.902
M ₃	0.432	-0.126	-0.076	-0.144	-0.427	-0.542	-0.347	-0.223	-0.214	-0.547	-0.649
M ₄	0.213	-0.021	-0.347	-0.431	-0.489	-0.643	-0.298	-0.347	-0.431	-0.489	-0.643
Extension superiority							-0.077	-0.286	-0.428	-0.627	-0.725

The overall results indicate that the current state of the coal mine is classified as “safe” (Grade I), showing that the proposed framework can effectively complete knowledge fusion and integrated classification for complex safety systems. More importantly, the model not only provides the final classification conclusion but also reveals differences in the contributions of different knowledge dimensions and knowledge attributes. In the personnel management dimension, employee safety assessment pass rate and violation operation frequency constitute the main weak knowledge links. The equipment dimension performs relatively well overall, but equipment integrity rate is close to the safety boundary. The operation environment dimension forms a significant constraint on the overall result, indicating that dynamic environment knowledge has high sensitivity in complex systems. In the management dimension, emergency drill frequency and safety investment proportion indicate that there is still room for improvement in systemic safeguarding knowledge. It can be seen that the framework has both classification and interpretation functions, and can provide a basis for subsequent knowledge intervention and governance decision-making.

It should be noted that this paper conducts empirical analysis using a single coal mine case. The purpose is not to restrict the proposed knowledge-driven evaluation framework to the coal mine safety evaluation scenario, but to carry out preliminary validation of the operability, stability, and applicable conditions of the framework by using a typical complex safety system with characteristics of multi-source information, dynamic disturbances, and multidimensional risks. From the perspective of method stability, the setting of classical domains and joint domains provides relatively clear classification boundaries for different knowledge attributes, and the correlation function can characterise the degree of deviation and closeness of knowledge attributes relative to the boundaries of various knowledge classifications, thereby avoiding, to a certain extent, classification mutation that may be caused by single-threshold judgement. Meanwhile, the introduction of knowledge importance weights enables the integrated superiority calculation to comprehensively reflect the differentiated influence of different knowledge attributes on system state determination, allowing the model to maintain a certain degree of result stability when facing local attribute fluctuations, and to respond to continuous deviation of key knowledge units. For knowledge attributes with dynamic variation characteristics, such as gas concentration mean, dust concentration compliance rate, and drainage system reliability, interval data and critical value reasoning are adopted in this paper, which helps reduce the interference of instantaneous fluctuations on evaluation results while retaining risk information under unfavourable working conditions. From the perspective of method applicability, the framework in this paper follows the general logic of “knowledge unit identification—knowledge structure construction—knowledge correlation reasoning—knowledge fusion classification”. Its core does not depend on specific empirical indicators of the coal mine industry, but relies on structured representation of knowledge attributes in complex systems, reasonable setting of classification boundaries, and effective characterisation of knowledge importance. Therefore, this method can provide a certain methodological reference for other complex systems characterised by multi-source heterogeneous knowledge, uncertain state boundaries, and inconsistent evaluation conclusions [18]. At the same time, the application effect of the framework is still affected by the rationality of classical domain, joint domain, and weight settings. Especially in cross-industry or cross-scenario applications, it is necessary to adapt knowledge boundaries and weight parameters by combining relevant domain standards, expert knowledge, and historical data. Subsequent studies can further introduce multi-case validation, weight sensitivity analysis, and dynamic boundary updating mechanisms to test and improve the transferability and robustness of the framework in different complex systems [19, 20].

5 Conclusion

This paper addressed the problems of multi-source knowledge conflict and evaluation inconsistency in complex safety systems, and proposed a knowledge-driven evaluation framework based on Extenics theory. The framework constructed a knowledge structure through conjugate analysis, transforming safety elements such as personnel, equipment, environment, and management into expressible knowledge units; knowledge representation was realised through matter-element models, classical domains, and joint domains; knowledge reasoning, knowledge fusion, and integrated classification were achieved through correlation functions and extension superiority. The coal mine case shows that the proposed method can not only provide safety classification results consistent with actual conditions, but also identify weak knowledge links and explain the boundary state of the system. Compared with traditional engineering-oriented evaluation descriptions, this paper further emphasised the theoretical significance of the method in knowledge modelling, knowledge conflict handling, and knowledge reasoning, and provided a scalable unified framework for risk cognition and evaluation research in complex systems.

Author Contributions

Conceptualization, H.L.; methodology, H.L. and Z.B.; complex system safety evaluation framework design, H.L.; knowledge representation and reasoning model construction, H.L. and Z.B.; validation and formal analysis, H.L.; investigation, Z.B.; resources, H.L.; data curation, Z.B.; writing—original draft preparation, H.L.; writing—review and editing, H.L. and Z.B.; visualization, Z.B.; supervision, H.L.; project administration, H.L.; funding acquisition, H.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflict of interest.

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